

L-17: Convergence Theorems

Monotone Convergence Theorem (MCT)

Let $\{X_n\}$ be a sequence of non-negative random variables (i.e., $X_n \geq 0$). If $X_n(\omega) \uparrow X(\omega)$ almost surely for all $\omega \in \Omega$, then

$$\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$$

The conditions ($X_n \geq 0$ and $X_n \uparrow X$) are very important. If $X_n \downarrow X$, the MCT may fail.

Example 1

Let $U \sim \text{Uniform}(0, 1)$ and define $X_n = n \cdot \mathbb{I}(U \leq 1/n)$.

- $X_n \geq 0$.
- $X_n \rightarrow 0$ almost surely. (This is a decreasing sequence).
- $\mathbb{E}[X_n] = n \cdot P(U \leq 1/n) = n \cdot (1/n) = 1$.
- Thus, $\lim \mathbb{E}[X_n] = 1 \neq 0 = \mathbb{E}[\lim X_n]$.

Example 2 (Cauchy)

Let Y be a Cauchy random variable. Define $X_n = |Y| \cdot \mathbb{I}(|Y| > n)$.

- $X_n \geq 0$.
- As $n \rightarrow \infty$, $\mathbb{I}(|Y| > n) \rightarrow 0$, so $X_n \rightarrow 0$ almost surely.
- This sequence is decreasing: $X_n \downarrow 0$.

Let's compute the expectation. The Cauchy PDF is $f_Y(y) = \frac{1}{\pi(1+y^2)}$.

$$\mathbb{E}[X_n] = \mathbb{E}[|Y| \mathbb{I}(|Y| > n)] = \int_{|y|>n} |y| \frac{1}{\pi(1+y^2)} dy$$

By symmetry, this is:

$$\mathbb{E}[X_n] = \frac{2}{\pi} \int_n^\infty \frac{y}{1+y^2} dy$$

Let $z = 1 + y^2$, so $dz = 2y dy$.

$$= \frac{1}{\pi} \int_{1+n^2}^\infty \frac{dz}{z} = \frac{1}{\pi} [\ln z]_{1+n^2}^\infty = \infty$$

So, $X_n \downarrow 0$ a.s., but $\mathbb{E}[X_n] = \infty$ for all n .

Example 3 (Modifying Ex. 2)

Define a new sequence $\hat{X}_n = -X_n = -|Y| \mathbb{I}(|Y| > n)$.

- $\hat{X}_n \leq 0$.
- Since $X_n \downarrow 0$, we have $\hat{X}_n \uparrow 0$ almost surely.
- $\mathbb{E}[X_n] = \infty \implies \mathbb{E}[\hat{X}_n] = -\infty$.
- Here, $\lim \mathbb{E}[\hat{X}_n] = -\infty \neq 0 = \mathbb{E}[\lim \hat{X}_n]$.

This shows the non-negativity condition in MCT is essential.

Question:

Suppose $Z_n \geq 0$ and $Z_n \downarrow Z$ almost surely. Does $\mathbb{E}[Z_n] \downarrow \mathbb{E}[Z]$?

- **Answer:** Yes, if $\mathbb{E}[Z_1] < \infty$.
- $0 \geq Z$, $\mathbb{E}[Z]$ is also defined
- Take $X_n = -Z_n \geq 0$ and $X = -Z$. Then $X_n \uparrow X$. Applying MCT (ignoring the non-negativity requirement) gives $\mathbb{E}[X_n] \uparrow \mathbb{E}[X] \implies \mathbb{E}[-Z_n] \uparrow \mathbb{E}[-Z] \implies -\mathbb{E}[Z_n] \uparrow -\mathbb{E}[Z]$, which implies $\mathbb{E}[Z_n] \downarrow \mathbb{E}[Z]$. (*This is the "reverse" version of MCT, which requires $\mathbb{E}[Z_1] < \infty$ to be valid.*)

Dominated Convergence Theorem (DCT)

Suppose $X_n \rightarrow X$ almost surely $\forall \omega \in \Omega$.

Further, assume there exists a random variable say $Y \geq 0$ such that

1. $\mathbb{E}[Y] < \infty$
2. $|X_n| \leq Y$ almost surely, for all $n \geq 1$.

Then $X_n \rightarrow X$ in L_1 (i.e., $\mathbb{E}[|X_n - X|] \rightarrow 0$). In particular,

$$\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$$

Observations

1. Under the DCT conditions, $X_n \in L_1$ because $\mathbb{E}[|X_n|] \leq \mathbb{E}[Y] < \infty$.
2. The conclusion $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$ follows trivially from $X_n \rightarrow X$ in L_1 , because

$$|\mathbb{E}[X_n] - \mathbb{E}[X]| = |\mathbb{E}[X_n - X]| \leq \mathbb{E}[|X_n - X|] \rightarrow 0$$

Corollary:

If $X_n \rightarrow X$ almost surely and there is a constant $M < \infty$ such that $|X_n| \leq M$ for all n , then

$$\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$$

(This is a special case of DCT, where we can choose $Y = M$, and $\mathbb{E}[Y] = M < \infty$.)

$$E[X_n] \rightarrow E[X].$$

Example: Let $U \sim \text{Uniform}(0, 1)$. Then $E[U^n] \rightarrow 0$.

Example 1:

$$\begin{aligned} X_n &= n I(U \leq \frac{1}{n}) \\ X_n &\rightarrow 0, \quad E[X_n] = 1 \end{aligned}$$

Example 2:

$$\begin{aligned} X_n &= |Y| I(|Y| > n) \\ X_n &\rightarrow 0, \quad E[X_n] = \infty \end{aligned}$$

Example 3:

$$\begin{aligned} X_n &= |Y| I(|Y| > n) \\ X_n &\rightarrow 0, \quad E[X_n] \rightarrow -\infty \end{aligned}$$

Monotone Approximation Theorem

If X is a non-negative random variable ($X \geq 0$), then there exists a sequence of **simple** random variables $X_n \geq 0$ (which take only finitely many values) such that

$$X_n(\omega) \uparrow X(\omega) \quad \text{for all } \omega$$

Towards the Proof

Let

$$X : \Omega \rightarrow [0, \infty), \quad \omega \mapsto X(\omega)$$

fix $\omega \in \Omega$.

Define

$$X_n(\omega) = \begin{cases} X(\omega), & \text{if } n > X(\omega), \\ n, & \text{if } n \leq X(\omega). \end{cases}$$

For example,

$$X_1^* = \begin{cases} \frac{n}{2}, \\ - \end{cases} \quad X_2^*, \dots$$

Algorithm proceeds by halving the interval each time we notice in which half we have the point and proceed to do the same process again and again until we reach very close to it.